

Awarding Body: Arden University (Berlin)

Programme Name: Programming for Data Analytics

Module Name (and Part if applicable): COM7024

Assessment Title: Data Analytics in Python

Student Number: 24169956

Tutor Name: Prof. Ahmed Hassan

Word Count: 3,266

Please refer to the Word Count Policy on your Module Page for guidance

Table of Contents

1. Introduction	3
2. Data Understanding & Methodology	3
2.1 Brief data understanding	3
2.2 Digital tools used	3
3. Data Pre-processing	4
3.1 Initial data quality check	4
3.2 Handling data types and invalid values	4
3.3 Handling missing data	6
3.4 Outlier detection and treatment	7
4. Exploratory Data Analysis (EDA)	10
4.1 Overview of the cleaned dataset	10
4.2 Distribution of key numerical variables	10
4.3 Distribution of the target variable	11
4.4 Categorical factors and career change	12
4.4.1 Gender	12
4.4.2 Education level	12
4.5 Job-related attitudes and career change	13
4.6 Correlation analysis	15
5. Discussion	17
6. Conclusion	18
7. References	19
8. Appendix A	20
9. Appendix B	23

Introduction:

In this assignment, I've used a career-related dataset (**Career data_PDA_4053.xlsx**) for the **Programming for Data Analytics (COM7024)** module. The dataset contains information on **5,000** individuals and includes **21** variables that capture occupational traits, career-related factors, and demographics features for 5,000 individuals. **Field of study, current occupation, age, gender, years of experience, education level, industry growth rate, job satisfaction, work-life balance, job opportunities, and salary** are a few examples of variables. **Career Change Interest, Skills Gap, Certifications, Freelance Experience, Geographic Mobility, Professional Networks, Career Change Events, and Technology Adoption** are just a few of the variables that are recorded in the data that may have an impact on career mobility. **"Likely to Change Occupation"**, a binary indicator that indicates whether each person is thought to be likely to change their occupation, is the primary outcome variable. This analysis's main goal is to preprocess the dataset and investigate the relationship between these many variables and the probability of changing careers.

Data Understanding & Methodology

2.1 Brief data understanding

There were both numerical and categorical variables in the dataset, according to the first **Python** examination. **Age, years of experience, pay, job satisfaction, and work-life balance** scores are examples of **numerical features**; **gender, education level, field of study, current occupation, and industry growth rate** are examples of **categorical features**. The dataset is appropriate for classification tasks in the future because the goal variable, **"Likely to Change Occupation"**, is stored as a binary indicator (**0 = not likely, 1 = likely**).

2.2 Digital tools used

All analysis was done using **Python** on **Google Colab**. Colab was selected because it offers a ready-to-use environment that doesn't require installation and allows for notebook-style analysis. **Pandas** for data loading and preprocessing, **NumPy** for numerical calculations, and **matplotlib** and **seaborn** for visualization were the primary **libraries** utilized. I was able to examine data types, compute descriptive statistics, handle missing values, and create plots such as **correlation heatmaps, boxplots, and histograms**. I was able to work methodically with the dataset and record every stage of the research in code by using these digital tools.

Data Pre-processing

3.1 Initial data quality check

I used Python to perform a series of fundamental data quality tests before implementing any changes. I looked at the dataset's structure, the ranges of each variable, and the quantity of **missing data** using functions like **df.info()**, **df.describe()**, and **df.isna().sum()**. In addition to confirming that the dataset contained both **numerical and categorical variables**, these checks also identified several problems with the raw file. A few observations had unexpected codes in binary fields, and certain variables that ought to be numerical, such as **salary** and some **0/1** indicators, were recorded as **object (text)** types. Furthermore, several columns included **missing data**, and the boxplots and descriptive statistics for salary indicated the existence of **extreme outliers**. As illustrated in **Figure 3.1**, it displays numerous columns with nan values and object types.

```
*** <class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 21 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Field of Study       4999 non-null   object
1   Current Occupation   4996 non-null   object
2   Age                  4996 non-null   float64
3   Gender                4993 non-null   object
4   Years of Experience   4998 non-null   float64
5   Education Level       4994 non-null   object
6   Industry Growth Rate  4995 non-null   object
7   Job Satisfaction      4998 non-null   float64
8   Work-Life Balance     4997 non-null   float64
9   Job Opportunities     4993 non-null   float64
10  Salary                4994 non-null   object
11  Job Security          4996 non-null   float64
12  Career Change Interest 4995 non-null   object
13  Skills Gap            4995 non-null   float64
14  Certifications        4995 non-null   object
15  Freelancing Experience 4994 non-null   float64
16  Geographic Mobility    4996 non-null   object
17  Professional Networks 4994 non-null   float64
18  Career Change Events  4998 non-null   float64
19  Technology Adoption   4996 non-null   float64
20  Likely to Change Occupation 4999 non-null   float64
dtypes: float64(12), object(9)
memory usage: 820.4+ KB

*** Column: Field of Study
['Medicine' 'Education' 'Arts' 'Computer Science' 'Business'
'Mechanical Engineering' 'Biology' 'Law' 'Economics' 'Psychology' nan]

Column: Current Occupation
['Business Analyst' 'Economist' 'Biologist' 'Doctor' 'Lawyer'
'Software Developer' 'Artist' 'Psychologist' 'Teacher'
'Mechanical Engineer' nan]

Column: Gender
['Male' 'Female' nan]

Column: Education Level
['High School' 'Master's' 'PhD' 'Bachelor's' nan]

Column: Industry Growth Rate
['High' 'Low' nan 'Medium']

Column: Salary
[198266 96803 65920 ... 115699 182137 158505]

Column: Career Change Interest
[0 1 nan 'A']

Column: Certifications
[0 1 nan 'A']

Column: Geographic Mobility
[1 0 nan 'A']
```

Figure 3.1: Initial structure and sample categories of the career dataset in Python.

Early detection of these issues made it easier for me to organize the preprocessing stages that followed, which concentrated on fixing data types, handling outliers before proceeding to **exploratory analysis**.

3.2 Handling data types and invalid values

Following my discovery of these problems, I initially concentrated on fixing the data types of various variables. Salary and a few other number fields were saved as **objects (text)** in the raw dataset instead of numerical values. I transformed Salary into a correct numerical column using **pandas' "to_numeric"** function. This adjustment was crucial because it made it possible for me to compute significant descriptive statistics, create boxplots and histograms for salary, and incorporate them into the **correlation study**. Additionally, I

verified that other score-based variables, like **work-life balance** and **job satisfaction**, were kept in the proper numerical format.

Next, I cleaned several binary variables, such as **Geographic Mobility**, **Certifications**, and **Career Change Interest**. These columns included the expected **0** and **1** numbers as well as unexpected categories classified as **"A"**, according to an initial review. I considered this code as an invalid entry and substituted missing values (**NaN**) because it did not fit the binary structure and could not be safely interpreted. In order to make these variables consistent and usable for further analysis, I then changed them to a numerical **0/1** format.

Lastly, I made sure that the goal variable, **"Likely to Change Occupation"**, was kept as an integer indicator, where **1** meant **"likely"** and **0** meant **"not likely"**. This standardization is crucial for descriptive analysis as well as any classification models that may be developed in the future using this dataset. As shown in **Table 3.1 & Figure 3.2**.

Variable	Original type/issue	Cleaned type/action
Salary	Object (text)	Converted to Numeric using to_numeric
Career Change Interest	Object, values 0, 1, 'A'	'A' → NaN, then converted to numeric 0/1
Certifications	Object, values 0, 1, 'A'	'A' → NaN, then converted to numeric 0/1
Geographic Mobility	Object, values 0, 1, 'A'	'A' → NaN, then converted to numeric 0/1
Likely to Change Occupation	Object / Float	Converted to Integer 0/1

Table 3.1: Summary of key variables and data-type cleaning actions.

```

Data columns (total 21 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Field of Study                        4999 non-null   object
1   Current Occupation                    4996 non-null   object
2   Age                                  4996 non-null   float64
3   Gender                               4993 non-null   object
4   Years of Experience                   4998 non-null   float64
5   Education Level                       4994 non-null   object
6   Industry Growth Rate                  4995 non-null   object
7   Job Satisfaction                      4998 non-null   float64
8   Work-Life Balance                     4997 non-null   float64
9   Job Opportunities                     4993 non-null   float64
10  Salary                               4993 non-null   float64
11  Job Security                          4996 non-null   float64
12  Career Change Interest                4994 non-null   float64
13  Skills Gap                            4995 non-null   float64
14  Certifications                        4993 non-null   float64
15  Freelancing Experience                 4994 non-null   float64
16  Geographic Mobility                    4995 non-null   float64
17  Professional Networks                  4994 non-null   float64
18  Career Change Events                  4998 non-null   float64
19  Technology Adoption                   4996 non-null   float64
20  Likely to Change Occupation            4999 non-null   int64
dtypes: float64(15), int64(1), object(5)

```

Figure 3.2: Data types and non-null counts after cleaning the dataset.

3.3 Handling missing data

To assess the amount of missing data in the dataset, I used Python's `isna().sum()` function to count the number of null values in each column. **Figure 3.3's** left side displays the findings. A few variables, such as **Current Occupation**, **Age**, **Gender**, **Education Level**, **Industry Growth rate**, **Job Satisfaction**, **Work-Life Balance**, **Job Opportunities**, **Salary**, and **Career Change Interest**, had a few missing items. The percentage of missing data for each feature was generally modest, ranging from **one to six out of 5,000** observations.

Field of Study	0	Field of Study	0
Current Occupation	3	Current Occupation	0
Age	3	Age	0
Gender	6	Gender	0
Years of Experience	1	Years of Experience	0
Education Level	5	Education Level	0
Industry Growth Rate	4	Industry Growth Rate	0
Job Satisfaction	1	Job Satisfaction	0
Work-Life Balance	2	Work-Life Balance	0
Job Opportunities	6	Job Opportunities	0
Salary	6	Salary	0
Job Security	3	Job Security	0
Career Change Interest	5	Career Change Interest	0
Skills Gap	4	Skills Gap	0
Certifications	6	Certifications	0
Freelancing Experience	5	Freelancing Experience	0
Geographic Mobility	4	Geographic Mobility	0
Professional Networks	5	Professional Networks	0
Career Change Events	1	Career Change Events	0
Technology Adoption	3	Technology Adoption	0
Likely to Change Occupation	0	Likely to Change Occupation	0

Figure 3.3:

Number of missing values per variable before imputation (left) and after imputation (right).

Rather than deleting rows from the dataset, I chose to keep all records and replace the missing values because the amount of missingness was small. For both numerical and categorical data, I used several imputation techniques. I used the **median** value of each

column to fill in the blanks for numerical characteristics like **Age, Years of Experience, Salary, Job Satisfaction, and Work-Life Balance**. Because it is less sensitive to extreme values – which is crucial for variables like **salary** where outliers were present – the **median** was selected over the **mean**. To maintain the current distribution categories for categorical variables like **gender, education level, field of study, and current occupation**, I substituted the mode – the most frequent category—for missing values.

After imputation, I've used `isna().sum()` to double-check the dataset and verified that there were no missing values in any of the variables. This method ensured the descriptive statistics and visualizations were based on a complete dataset while maintaining all 5,000 original records.

3.4 Outlier detection and treatment

Plots and descriptive statistics for **Salary** revealed the existence of **extreme outliers** during the first investigation. The greatest salary value was well above the **median** and many times bigger than the average range, suggesting that the distribution's general shape and **mean** could be significantly distorted by a small number of observations. Using the raw data, I created **boxplots and salary histograms** to better explore this.

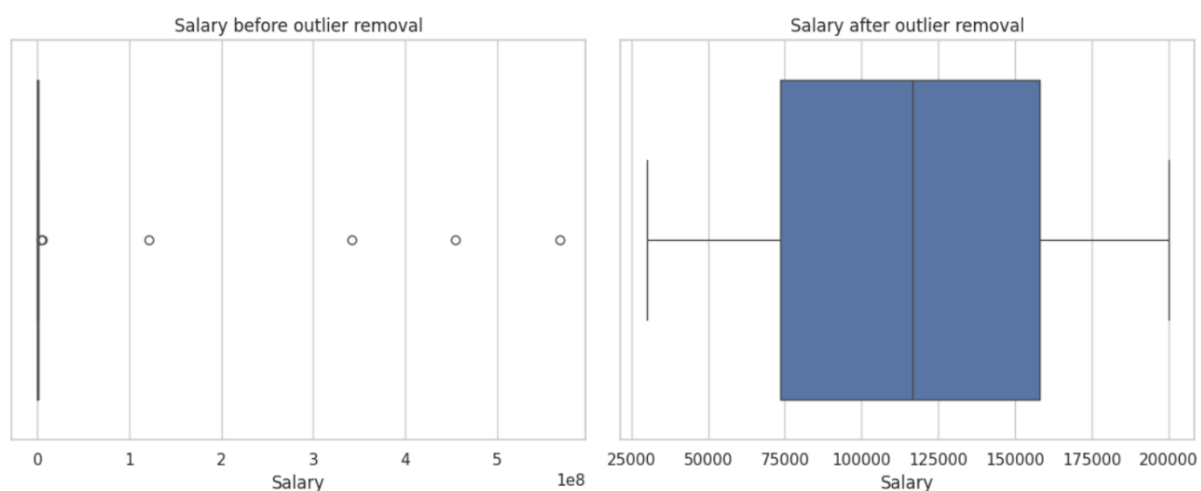


Figure 3.4: Boxplots of Salary before (left) and after (right) outlier removal.

As shown in **Figure 3.4 (left)**, the original boxplot is dominated by several extremely high salary values, which appear as isolated points far to the right of the main distribution.

To deal with these extreme results in a systematic approach, I employed the **interquartile range (IQR) method**. I first determined the **salary's** first and third quartiles (**Q1 and Q3**), and then I calculated the **IQR as $Q3 - Q1$** . Any observations with a salary **below $Q1 - 1.5 \times IQR$** or **above $Q3 + 1.5 \times IQR$** were considered outliers. The remaining observations were kept as the cleaned Salary values, and all rows that fell outside of these bounds were eliminated from the dataset. With this method, the distribution is not randomly trimmed; instead, only the most **extreme examples are eliminated**.

“ In total, only 6 observations (out of 5,000) were removed as Salary outliers, so the sample size remained almost unchanged.”

Interquartile Range (IQR)	
Lower bound	-53446.75
Upper bound	285167.25
No. of Salary outliers	06

Table 3.2: IQR findings

The effect of this procedure can be seen clearly in **(Figures 3.4 & 3.5)**. Because the extreme values stretch the x-axis, nearly all incomes are compressed into a small band close to zero before cleaning **(Figure 3.5, left)**. The existence of extremely big outliers is confirmed by the original boxplot in **Figure 3.4**, which also displays several points far to the right of the main box. The cleaned **boxplot (Figure 3.4, right)** shows a much more condensed and comprehensible range once the IQR rule is applied, and the histogram **(Figure 3.5, right)** exhibits a distribution that is still **right-skewed** but is now focused inside a reasonable pay interval.

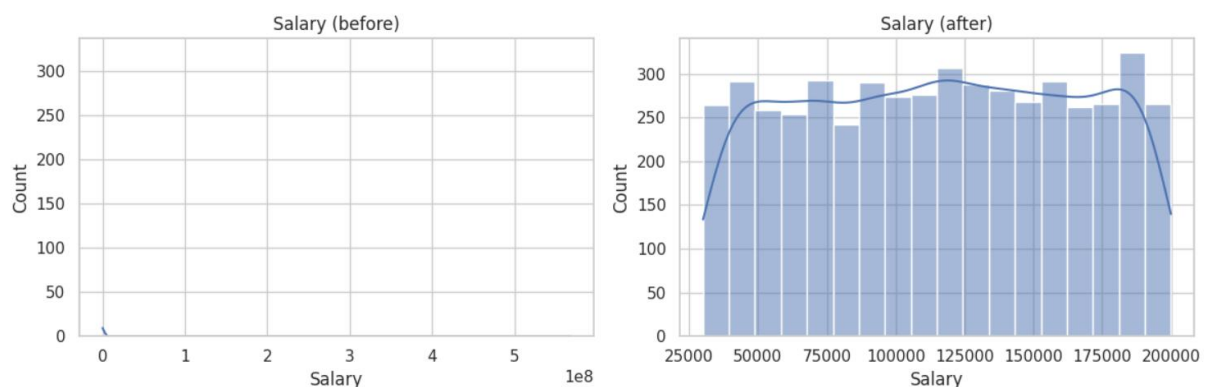


Figure 3.5: Histograms of Salary before (left) and after (right) outlier removal.

The primary summary statistics for salary before and after outlier removal are contrasted in **Table 3.3**. The **mean salary** significantly drops and approaches the **median**, suggesting that a small number of erroneous values have less of an impact on the cleaned data.

Statistics	Salary (before)	Salary (after)
<i>Mean</i>	414,935	115,867
<i>Median</i>	116,424	116,424
<i>Std dev</i>	11,490,800	48,869
<i>Min</i>	30,010	30,016
<i>Max</i>	567,896,500	199,976
<i>Skewness</i>	41.94	-0.02

Table 3.3: Summary statistics for Salary before and after outlier removal

The distribution is still slightly **skewed to the right**, but the maximum salary falls to a far more reasonable level, and the **skewness decreases**. These modifications imply that the cleaned Salary variable provides a more reliable basis for further investigation.

Additionally, I compared the distributions of **years of experience and age** before and after using the outlier approach. The histograms of these variables were **almost unchanged**, indicating that the outlier handling primarily impacted inflated **salary numbers** rather than distorting other important characteristics.

Overall, the outlier treatment stage preserved the dataset's fundamental structure while creating a cleaner, more realistic **Salary** variable. The dataset is now prepared for **exploratory data analysis (EDA)**, as presented in the next section, with corrected data types, imputed missing values, and robust outlier handling.

Two factors led us to this conclusion. First, the high incomes are more compatible with data entry or coding errors than with true earnings because they were several orders of magnitude larger than the dataset's usual values. Second, the impact on sample size is minimal because only **6 out of 5,000 observations (0.12%)** were eliminated, but the advantages for lessening the distortion of the mean and scale of the salary distribution are significant. These factors led to the conclusion that IQR-based removal was a suitable and cautious preprocessing step.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (**EDA**) is a phase that occurs throughout the **early stages of the data science** life cycle and entails conducting analysis over the data. Individuals become familiar with the source data after running EDA techniques and become aware of any **irregularities or anomalies that might be present in the data**. Manually evaluating the data and performing all the EDA techniques might be **time-consuming** and tedious. Rather than building predictive models, the aim is to **use summary statistics and visualisations** to identify trends, check assumptions, and generate insights about the data.

In this assignment, **EDA** is used to examine how **demographics, job-related, and career-related variables** have an effect after preprocessing, and to see how they are going to relate to the target variable **“Likely to Change Occupation”**. The distribution of important characteristics is shown in the following subsections, which also look at the aspect that seems to be most closely linked to the chance of **changing careers**.

4.1 Overview of the cleaned dataset

I conducted an exploratory analysis using the cleaned dataset (**4,994 observations and 21 variables**) following the preprocessing outlined in **Section 3**. At this point, all features had the proper data types, there were no more missing values, and the IQR technique had eliminated severe salary outliers. This ensured that the relationships between variables, summary statistics, and visualizations were founded on reliable and consistent values. **Age and Years of Experience** distributions hardly changed after outlier handling, as shown in **Figure 4.1**. The distribution of important factors and their relationship to the probability of changing occupations are examined in the following subsections.

4.2 Distribution of key numerical variables

The distribution of **Age and Years of Experience** is shown in **Figure 4.1**. Both variables are spread across a wide range of values, with Age roughly **covering 21 – 59 years** and **Years of Experience ranging from 0 – 39 years**. The nearly identical appearance of the before and after plots indicates that Salary was the primary variable changed by the outlier treatment and that these variables were not distorted.

The cleaned salary distribution (**Figure 3.5, right**) indicates that most people make between **30,000 and 200,000 annually**. Although there are no longer any ridiculous extreme values dominating the scale, the curve is somewhat tilted to the right, suggesting that higher salaries are less usual. This gives a more accurate picture of the sample’s **average income levels**.

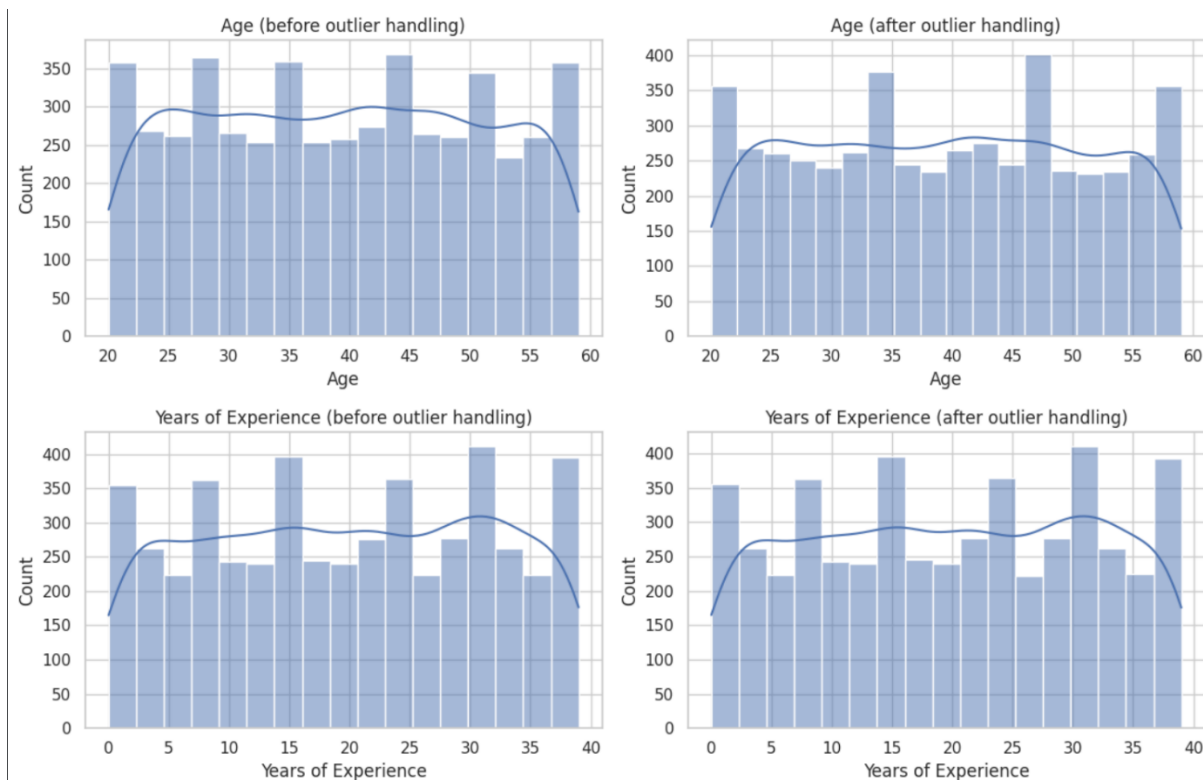


Figure 4.1: Distributions of Age and Years of Experience before (left) and after (right) outlier handling.

4.3 Distribution of the target variable

The distribution of the goal variable “**Likely to Change Occupation**” is displayed in **Figure 4.2**. Out of the **4,993** people in the cleaned dataset, **2,143 (42.9%)** are classified as **Not likely (0)** and **2,850 (57.1%)** as **Likely (1)** to change career. As a result, there is a minor imbalance in the classes, with a greater percentage of prospective job changers. A straightforward model that consistently forecasts “**Likely to change**” would already attain an accuracy of **57%**, so while this isn’t severe, it is crucial to bear in mind for any future modelling.

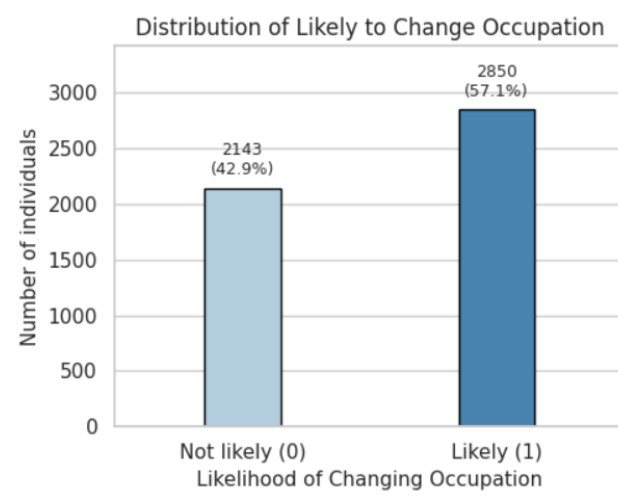


Figure 4.2: Distribution of Likely to Change Occupation (0 vs 1 bar chart with counts & %)

4.4 Categorical factors and career change

4.4.1 Gender

Figure 4.3 illustrates how similar the percentages of respondents who are likely to change occupation are for men and women (**about 56-58% in both groups**). This implies that career-change aspirations are not significantly differentiated by gender in this group. Therefore, adding gender as a key feature on its own is unlikely to significantly increase the accuracy of any prediction model that employs this information.

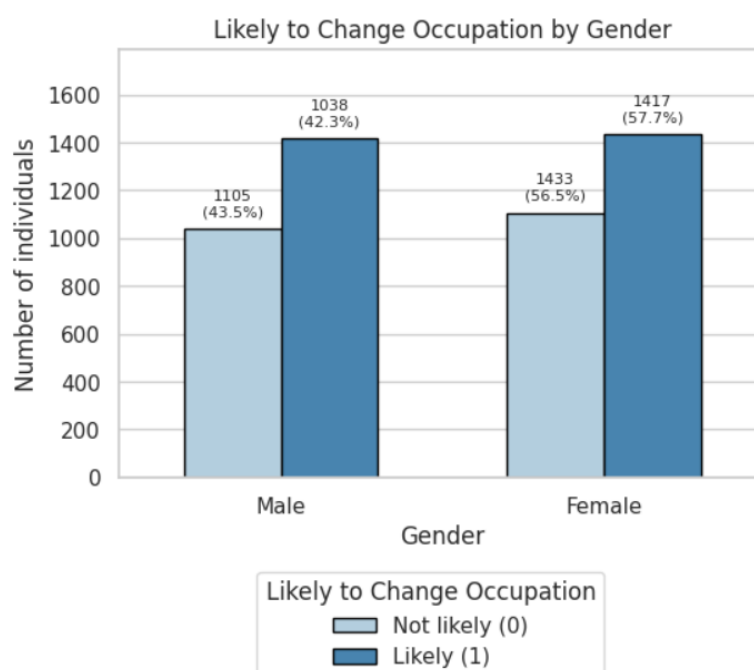


Figure 4.3: Likely to Change Occupation by Gender (grouped bar chart with percentages).

4.4.2 Education level

Figure 4.4 shows the distribution of the target variable across education levels. The percentage of people who are **likely to change** careers for each category – **High School, Bachelor's, Master's, and PhD** – all hover around **50% (approximately 49-51% in each group)**. Increased or lower education does not appear to be associated with a significantly increased likelihood of changing careers. This suggests that traits that lead to a career change exist at all levels of education and are likely influenced more by **ideas and work experiences than by education alone**.

Figure 4.4 indicates that the share of respondents likely to change occupation is approximately **50%** across all education levels. This suggests that having more education does not always prevent or promote the desire to change careers. Rather, it seems that variables other than formal schooling, such as job satisfaction or attitudes toward work, are what motivate people to change careers.

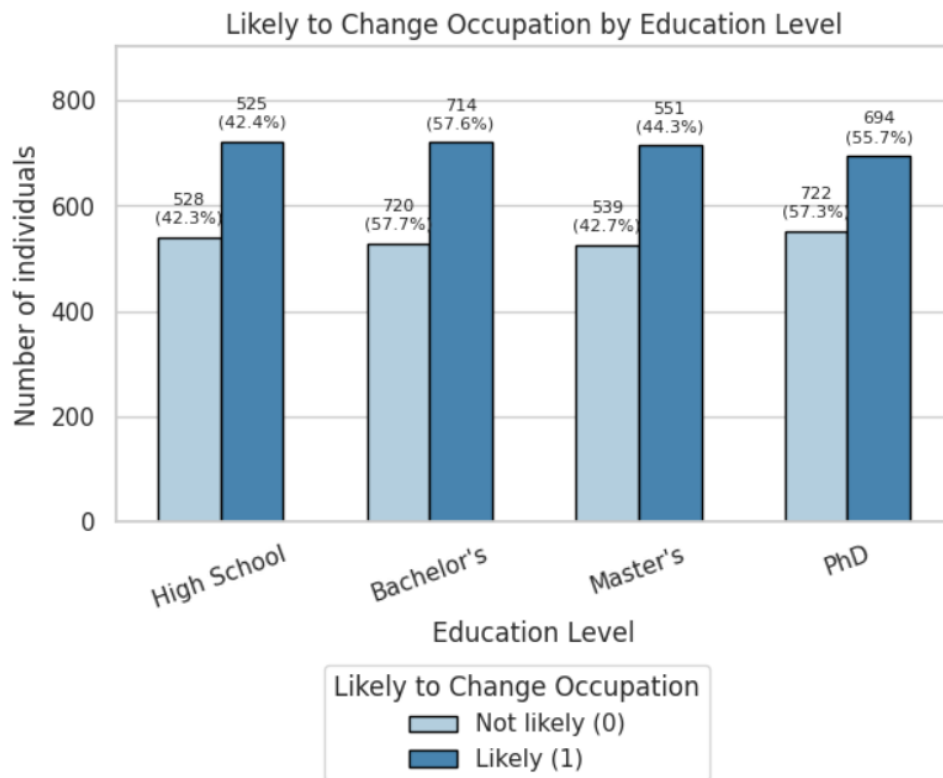


Figure 4.4: Likely to Change Occupation by Education Level (grouped bar chart).

4.5 Job-related attitudes and career change

I've compared **mean** scores for several variables between the two target groups (**Table 4.1**) and used a **boxplot** to visualize work satisfaction (**Figure 4.5**) to investigate which job-related aspects are most strongly associated with the possibility of changing occupations. The group means reveal that **age and years of experience** are nearly the same for those who are inclined to change careers and those who aren't (**both around 39 years and 19 – 20 years of experience**), suggesting that demographic traits alone do not significantly influence intentions to change careers.

<i>Likely to Change Occupation</i>	<i>Age</i>	<i>Years of Experience</i>	<i>Salary</i>	<i>Job Satisfaction</i>	<i>Work-Life Balance</i>	<i>Skills Gap</i>	<i>Career Change Events</i>
<i>0</i>	39.35	19.66	125562.68	7.49	5.58	5.61	1.01
<i>1</i>	39.37	19.83	108577.27	3.99	5.47	5.47	0.98

Table 4.1: Group means table (Age, Years of Experience, Salary, Job Satisfaction, Work-Life Balance, Skills Gap, Career Change Events) by target.

On the other hand, there are significant differences in job satisfaction between the two groups. On a scale of 1–10, people who are **unlikely to change careers** report an average level of satisfaction of roughly **7.5**, while those who are **likely to do** so report an average of **4.0**. This trend is supported by the **boxplot in Figure 4.5**, which shows that the “**likely**” group has a **median** satisfaction of about **3**, with scores concentrated in the lower half of the scale, while the “**not likely**” group has a **median** contentment of about **7**, with most scores falling between **6 & 9**. This suggests that job satisfaction and the probability of changing careers are strongly **negatively correlated**.

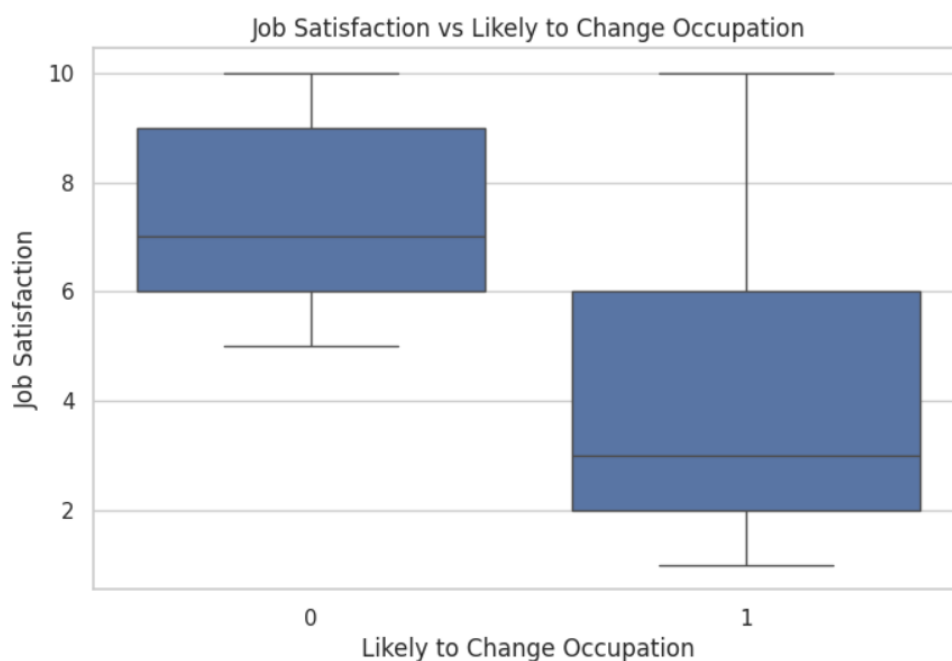


Figure 4.5: The Job Satisfaction boxplot vs Likely to Change Occupation.

Other variables show smaller differences. Although the effect is less noticeable than for satisfaction, the average salary for people who are **likely to change** careers is somewhat lower (**approximately 108,000 vs 126,000**), indicating that lower income may be a factor. The means of **Work-Life Balance**, **Skills Gap**, and **Career Change Events** are quite similar between the two groups, suggesting that they might have less of an impact when taken separately.

4.6 Correlation analysis

The correlation matrices for the numerical variables before and after data cleaning are shown in **Figures 4.6 & 4.7**. The general pattern is remarkably similar in **both versions**, indicating that the relationship between variables was not significantly distorted by the preprocessing procedures. Most correlations between predictors are **near zero**, which suggests that the variables capture distinct aspects of participants' circumstances and indicates minimal **multicollinearity**.

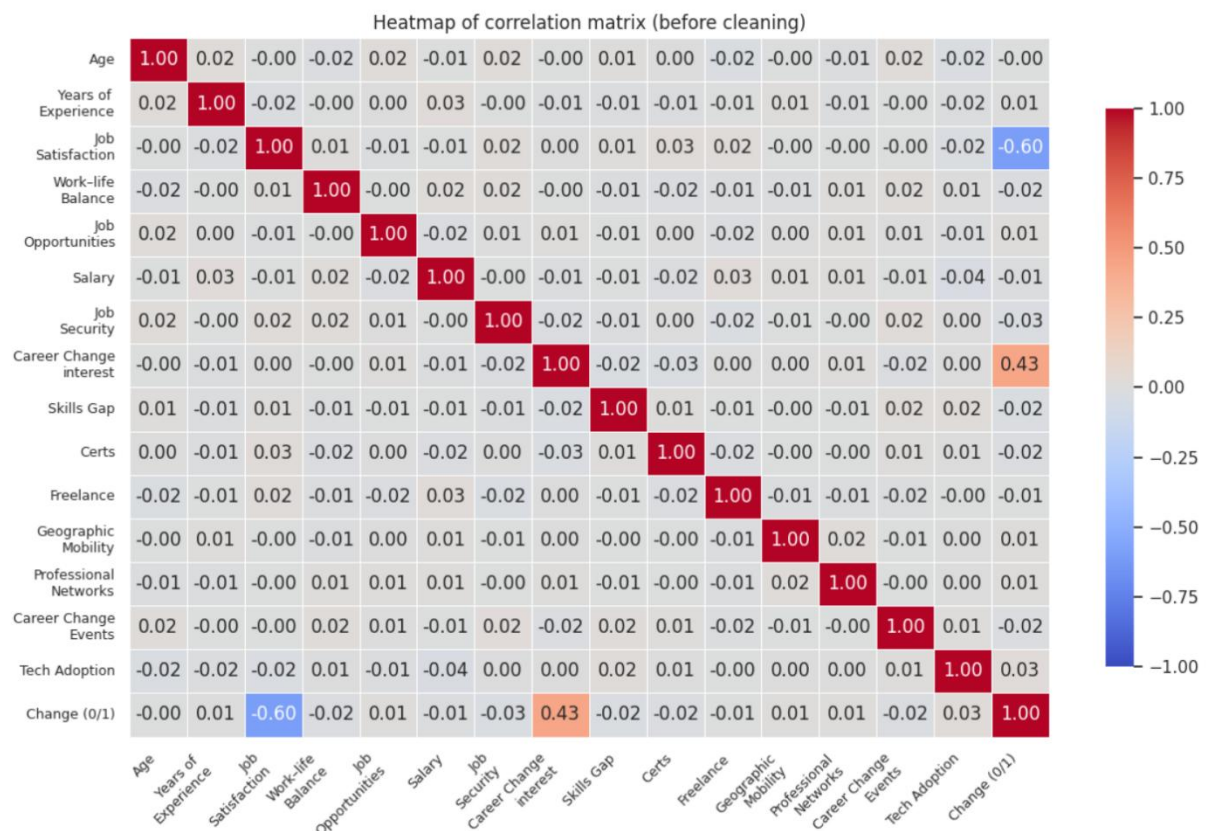


Figure 4.6: Heatmap of correlation matrix (before cleaning).

There are two noteworthy correlations with the goal of the variable. Higher job satisfaction is linked to a significantly reduced chance of changing careers, as evidenced by the large **negative correlation** between **job satisfaction** and **“Change (0/1)”** (roughly **-0.60**). On the other hand, **Career Change Interest** shows a **positive correlation of roughly 0.43**, meaning that those who show a greater desire to change occupations are more likely to be categorized as potential career changers. Following cleaning, unreasonable salary outliers are eliminated, resulting in a somewhat greater **negative correlation between Salary and the goal (around -0.17)**. All other correlations with the aim are low, implying that attitudinal variables (**satisfaction and interest**) are more relevant than demographic or structural factors.

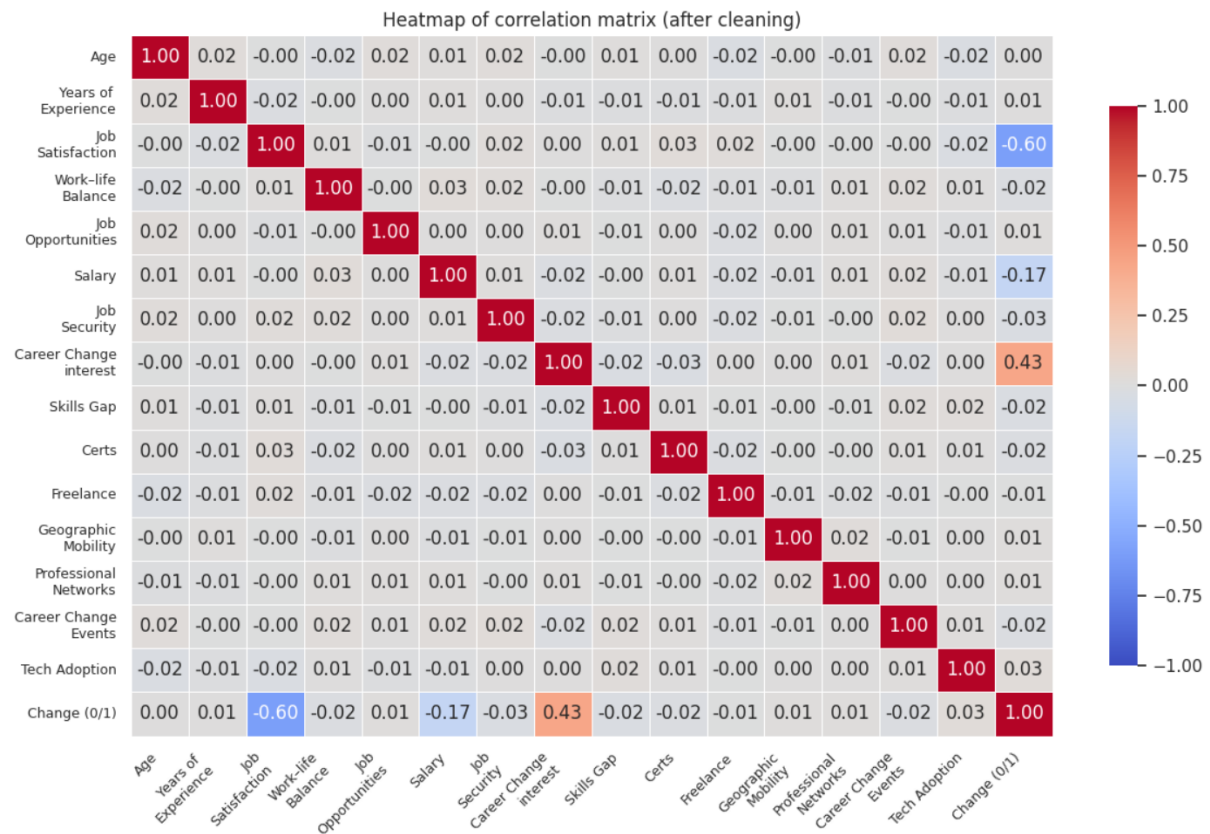


Figure 4.7: Heatmap of correlation matrix (after cleaning).

Overall, the **Exploratory Analysis (EDA)** indicates that stated interest in change and work satisfaction are more important factors influencing **career-change** intentions in this dataset than **age, experience, education, or gender**.

Discussion

The analysis shows that just over half of respondents (**around 57%**) are classified as **likely to change** occupation. The **mean** values for **years of experience and age** are nearly the same for the two target classes, and this trend is impressively stable across **gender and educational** groupings. Therefore, in this dataset, demographic factors seem to have very little bearing on plans to **change careers**.

In contrast, job-related attitudes are much more informative. Those who are **likely to change** careers have significantly **lower job satisfaction**, and there is a strong **positive correlation between the aim and career change interest**. This is verified using correlation analysis: **Career Change Interest** has a somewhat positive association with **“Change (0/1)”**; however, **Job Satisfaction** has a substantial **negative correlation**. After excluding extreme outliers, **salary** has a weaker negative association, indicating that while income may contribute, but is not the primary driver.

These findings have practical implications. Career services should concentrate on finding individuals who have a high interest in **changing occupations or low job satisfaction** and provide them with tailored advice. The cross-sectional, self-reported character of the study and the fact that some factors are conceptually close to the target, which means the results show association rather than causation, are its limitations.

Conclusion

This study showed how a real-world career dataset can be pre-processed and analysed using Python and Google Colab, progressing methodically from data type checks and missing-value imputation to outlier handling and exploratory visualization. A consistent dataset of 4,994 observations that was appropriate for additional statistical modelling was produced by the cleaning procedures (type corrections, conservative imputation, and IQR-based outlier elimination).

According to the analysis, there is little correlation between the target variable “Likely to Change Occupation” and demographic variables, including age, gender, years of experience, and education level. On the other hand, there are substantially stronger correlations between the outcome and attitudinal variables, especially work satisfaction and indicated interest in changing careers. This implies that people’s intentions to change careers are more influenced by their job satisfaction than by their identity or formal credentials.

These results have several real-world applications. To provide tailored support, such as coaching, internal mobility possibilities, or upskilling chances, career-advice services or HR teams employing comparable data should concentrate on identifying individuals with low job satisfaction or strong declared interest in change. Therefore, monitoring satisfaction levels in conjunction with basic demographics may offer an early warning system for possible employee turnover.

By using classification models (such as logistic regression or tree-based techniques) to forecast career-change intentions and comparing model performance with and without the attitudinal factors, future research could expand on this approach. Replicating the analysis on different datasets or over time would also be helpful to determine whether these trends hold under various labour market circumstances.

References

- 1) Pitroda, H. (2022) 'A Proposal of an Interactive Web Application Tool QuickViz: To Automate Exploratory Data Analysis', *2022 IEEE 7th International Conference for Convergence in Technology (I2CT)*. [Online]. Available at: <https://research.ebsco.com/> (Accessed: 10 November 2025).
- 2) Pandas (n.d.) *pandas.DataFrame.describe — Pandas documentation*. [Online]. Available at: <https://pandas.pydata.org/pandas.DataFrame> (Accessed: 10 November 2025).
- 3) Pandas (n.d.) *pandas.to_numeric — Pandas documentation*. [Online]. Available at: https://pandas.pydata.org/pandas.to_numeric (Accessed: 10 November 2025).
- 4) Pandas (n.d.) *pandas.isna — Pandas documentation*. [Online]. Available at: <https://pandas.pydata.org/pandas.isna> (Accessed: 10 November 2025).
- 5) Seaborn (n.d.) *seaborn.countplot*. [Online]. Available at: <https://seaborn.pydata.org/seaborn.countplot> (Accessed: 10 November 2025).
- 6) Seaborn (n.d.) *seaborn.histplot — Seaborn documentation*. [Online]. Available at: <https://seaborn.pydata.org/seaborn.histplot> (Accessed: 10 November 2025).
- 7) Seaborn (n.d.) *seaborn.boxplot — Seaborn documentation*. [Online]. Available at: <https://seaborn.pydata.org/seaborn.boxplot> (Accessed: 10 November 2025).
- 8) Seaborn (n.d.) *seaborn.heatmap — Seaborn documentation*. [Online]. Available at: <https://seaborn.pydata.org/seaborn.heatmap> (Accessed: 10 November 2025).
- 9) Matplotlib (n.d.) *Pyplot — API overview*. [Online]. Available at: https://matplotlib.org/pyplot_summary (Accessed: 10 November 2025).
- 10) Matplotlib (n.d.) *matplotlib.pyplot.subplots — Matplotlib documentation*. [Online]. Available at: <https://matplotlib.org/matplotlib.pyplot.subplots> (Accessed: 10 November 2025).
- 11) NIST/SEMATECH (n.d.) *Box Plot — NIST/SEMATECH e-Handbook of Statistical Methods*. [Online]. Available at: <https://www.itl.nist.gov/boxplot> (Accessed: 10 November 2025).

Appendix A – Code listings

A.1 Data import & type fixes

```
# imports
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from scipy import stats

sns.set(style="whitegrid")
plt.rcParams["figure.figsize"] = (8, 5)

df = pd.read_excel("/content/Career_data_PDA_4053.xlsx")
df.shape
df.head()
df.info()
df.describe()
df.describe(include='object')
```

Figure A.1.1: Imports

```
df_raw = df.copy()

# Convert Salary to numeric
df_raw['Salary'] = pd.to_numeric(df_raw['Salary'], errors='coerce')

# Clean and convert binary-like columns
binary_obj_cols = ['Career Change Interest', 'Certifications', 'Geographic Mobility']

for col in binary_obj_cols:
    # Treat 'A' as invalid -> set to NaN
    df_raw[col] = df_raw[col].replace('A', np.nan)
    # Convert to numeric (0/1)
    df_raw[col] = pd.to_numeric(df_raw[col], errors='coerce')

df_raw = df_raw.dropna(subset=['Likely to Change Occupation'])
df_raw['Likely to Change Occupation'] = df_raw['Likely to Change Occupation'].astype(int)

df_raw.info()
```

Figure A.1.2: Data Type fixes

A.2 Missing-value imputation (numeric: median; categorical: mode)

```
numeric_cols = df_raw.select_dtypes(include=['float64', 'int64']).columns
cat_cols = df_raw.select_dtypes(include=['object']).columns

df_raw[numeric_cols].describe()
df_raw[cat_cols].describe()
df_raw.isna().sum()
```

Figure A.2.1: Numeric and categorical before type fixing

```
df_clean = df_raw.copy()

# Numeric and categorical after type fixing
num_cols = df_clean.select_dtypes(include=['float64', 'int64']).columns
cat_cols = df_clean.select_dtypes(include=['object']).columns

# numeric
for col in num_cols:
    df_clean[col] = df_clean[col].fillna(df_clean[col].median())

# categorical
for col in cat_cols:
    df_clean[col] = df_clean[col].fillna(df_clean[col].mode()[0])

df_clean.isna().sum()
```

Figure A.2.2: Numeric and categorical after type fixing

A.3 Outlier handling (IQR rule for Salary)

```
salary_q1 = df_clean['Salary'].quantile(0.25)
salary_q3 = df_clean['Salary'].quantile(0.75)
salary_iqr = salary_q3 - salary_q1

lower_bound = salary_q1 - 1.5 * salary_iqr
upper_bound = salary_q3 + 1.5 * salary_iqr

print("Lower bound:", lower_bound)
print("Upper bound:", upper_bound)

outliers = df_clean[(df_clean['Salary'] < lower_bound) | (df_clean['Salary'] > upper_bound)]
print("Number of salary outliers:", outliers.shape[0])

df_no_outliers = df_clean[
    (df_clean['Salary'] >= lower_bound) & (df_clean['Salary'] <= upper_bound)
].copy()
```

... Lower bound: -53446.75
Upper bound: 285167.25
Number of salary outliers: 6

Figure A.3.1: Outlier handling (IQR rule for Salary)

A.4 EDA Plot

```
sns.set_theme(style="whitegrid")
counts = df_final['Likely to Change Occupation'].value_counts().sort_index()
labels = ['Not likely (0)', 'Likely (1)']
total = counts.sum()
fig, ax = plt.subplots(figsize=(5, 4))
sns.barplot(
    x=labels,
    y=counts.values,
    palette="Blues",
    width=0.3,
    edgecolor="black",
    ax=ax)
ax.set_title("Distribution of Likely to Change Occupation", fontsize=12)
ax.set_xlabel("Likelihood of Changing Occupation", fontsize=11)
ax.set_ylabel("Number of individuals", fontsize=11)
max_count = counts.max()
ax.set_ylim(0, max_count * 1.2)
for i, count in enumerate(counts.values):
    percent = 100 * count / total
    ax.text(
        i,
        count + max_count * 0.03,
        f"{count}\n({percent:.1f}%)",
        ha="center",
        va="bottom",
        fontsize=9)
plt.tight_layout()
plt.show()
```

Figure A.4.1: Bar plot - Distribution of Likely to Change Occupation

Appendix B – GitHub

-  **Python version in Google Colab environment:** 3.12.12
-  **GitHub:** [mirzasameer2000/Programming_for_Data_Analytics_COM7024](https://github.com/mirzasameer2000/Programming_for_Data_Analytics_COM7024)
-  **Dataset:** Career data_PDA_4053.xlsx
-  **Dataset after cleaning:** career_data_cleaned.csv
-  **Colab notebook:** 24169956_Programming_for_Data_Analytics_COM7024_[4053].ipynb
-  **Project Documentation:** 24169956 - Programming for Data Analytics -COM7024
[4053].pdf